

Industrial Internship Report on "Banking Information System"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "**Banking Information System**". The main objective of this project was to develop a system that manages banking operations such as account creation, deposit, withdrawal, balance inquiry, and transaction management efficiently using Core Java.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.



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1 Preface

This report summarizes the work completed during my six-week Industrial Internship conducted by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd.

Internships play a very important role in career development as they provide real-world exposure, practical knowledge, and industry experience beyond classroom learning. This internship helped me understand how theoretical concepts are applied in real-life projects.

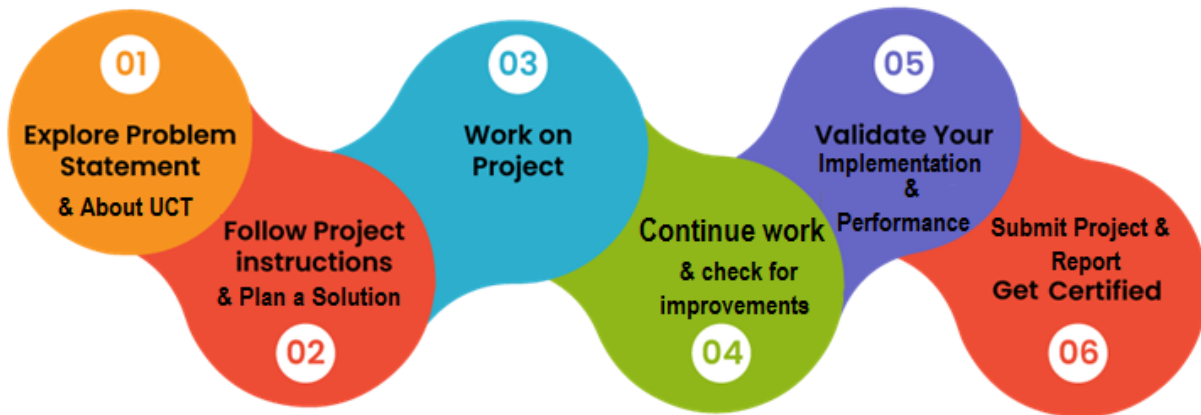
My project was **“Banking Information System using Core Java”**. The problem statement was to develop a system that can efficiently manage banking operations like account creation, deposit, withdrawal, and transaction tracking.

The program was well-structured and planned over six weeks, including learning sessions, project development, and final submission. I got the opportunity to learn new concepts, improve coding skills, and understand project development workflow.

During this internship, I learned Core Java concepts, problem-solving techniques, and how to build a complete application from scratch. It was a very valuable and enriching experience.

I would like to thank **Upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd**, and all mentors and coordinators who supported me throughout this internship.

How Program was planned



Your Learnings and overall experience.

Thank to all upskill campus , who have helped you directly or indirectly.

Message to juniors:

Always take internships seriously. Focus on learning, practice regularly, and try to build real projects. It will help you a lot in your future career.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



i. UCT IoT Platform ()

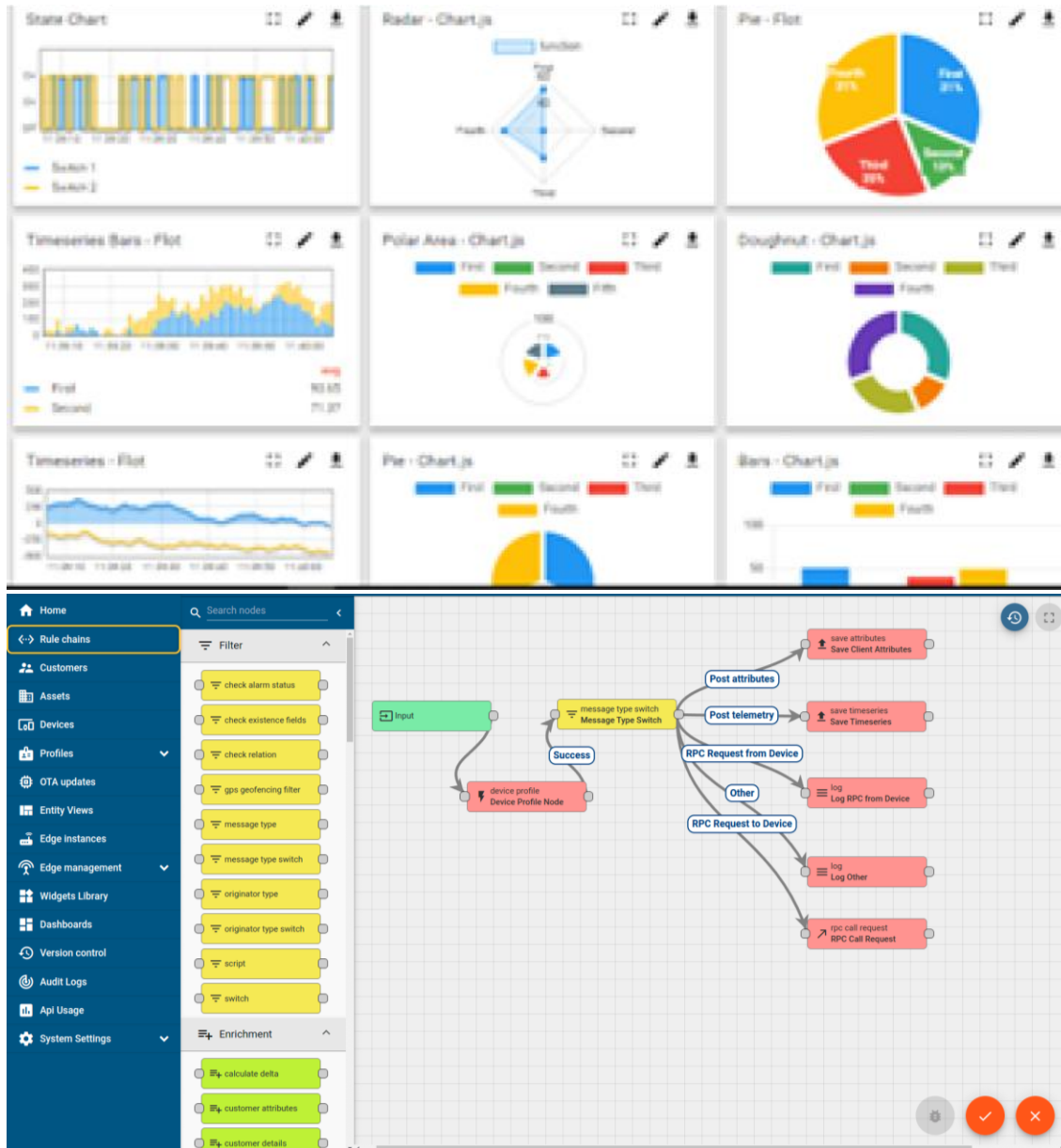
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



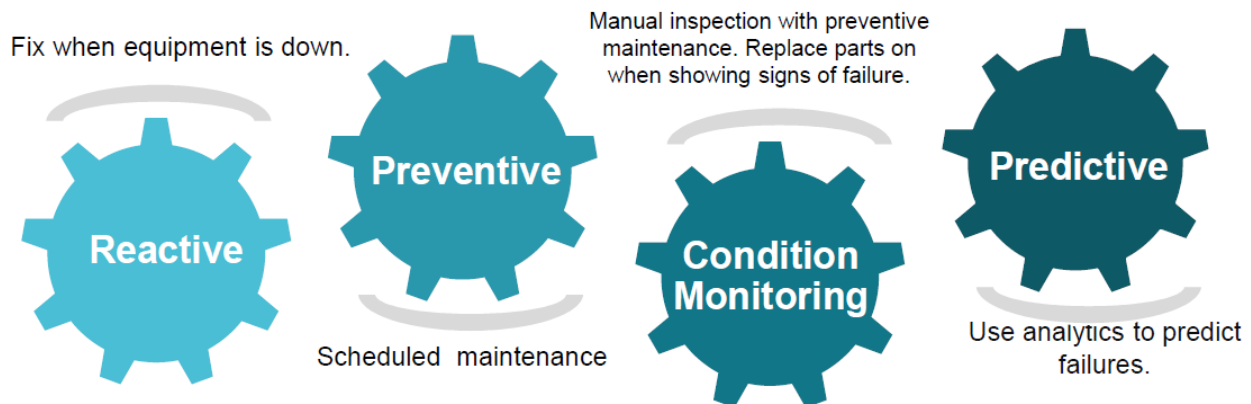


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

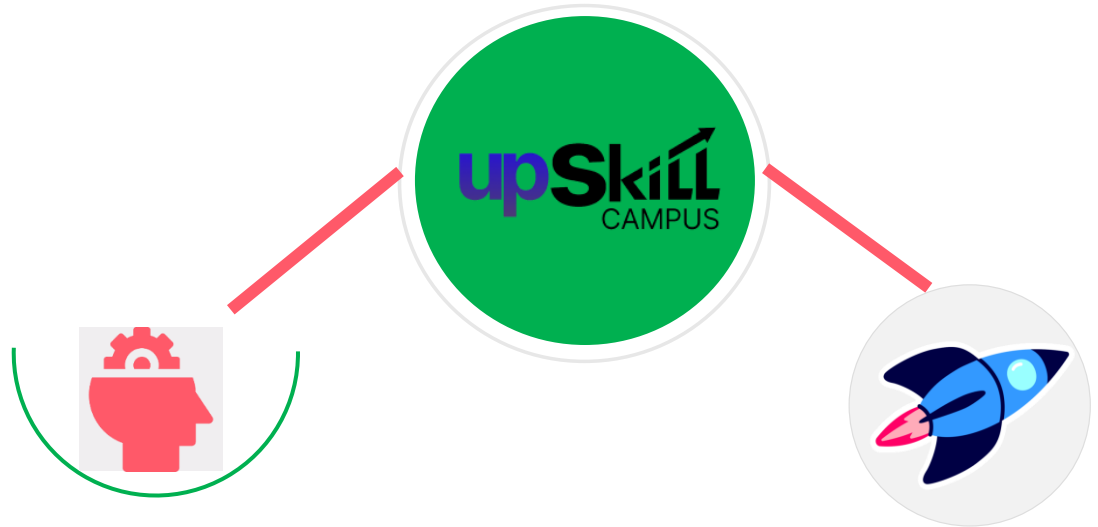
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

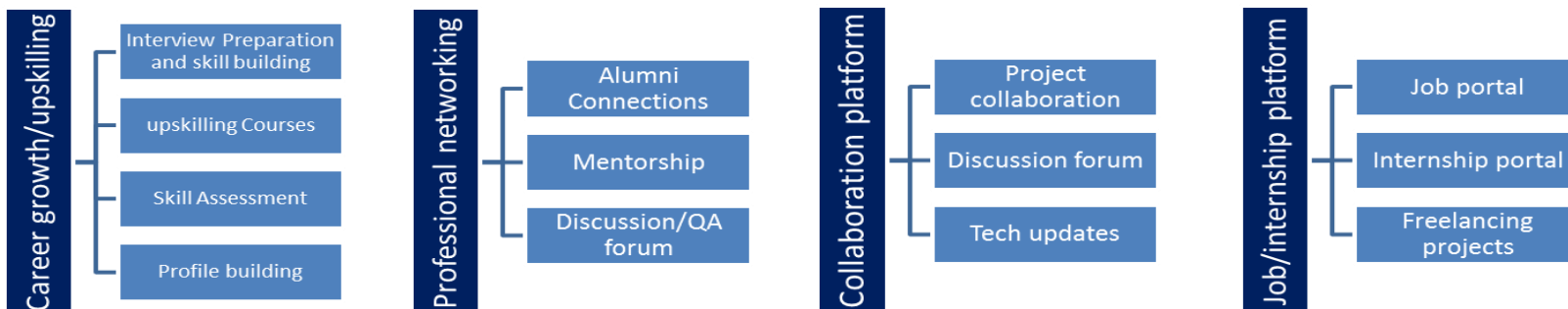
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1]

[2]

[3]

2.6 Glossary

Terms	Acronym

3 Problem Statement

The problem statement assigned during the internship was to develop a **Banking Information System** using Core Java.

The system should provide the following functionalities:

- User registration and login
- Account creation and management
- Deposit and withdrawal operations
- Fund transfer between accounts
- Transaction history and account statement

The aim was to design a system that simulates real banking operations in a structured and efficient manner using Java programming concepts.

4 Existing and Proposed solution

- **Existing Solutions:**

Existing banking systems are:

- Highly complex and large-scale
- Require advanced infrastructure
- Not suitable for beginners to understand

- **Proposed Solution:**

The proposed solution is a **console-based Banking Information System using Core Java** that:

- Simulates real banking functionalities
- Is simple and easy to use
- Works efficiently without database dependency
- **Value Addition:**
- Beginner-friendly design
- Demonstrates real-world banking workflow
- Helps in understanding OOP concepts practically

4.1 Code submission (Github link)

<https://github.com/rahulbag722136-netizen/Banking-Information-System-Java.git>

4.2 Report submission (Github link) :

<https://github.com/rahulbag722136-netizen/Banking-Information-System-Java.git>

5 Proposed Design/ Model

The system follows a **menu-driven architecture**:

Flow:

User → Menu → Operation → Processing → Output

Main Components:

- **User Class** → Stores user details
- **Account Class** → Manages account data
- **Transaction Class** → Stores transaction history
- **BankService Class** → Handles logic

5.1 High Level Diagram (if applicable)

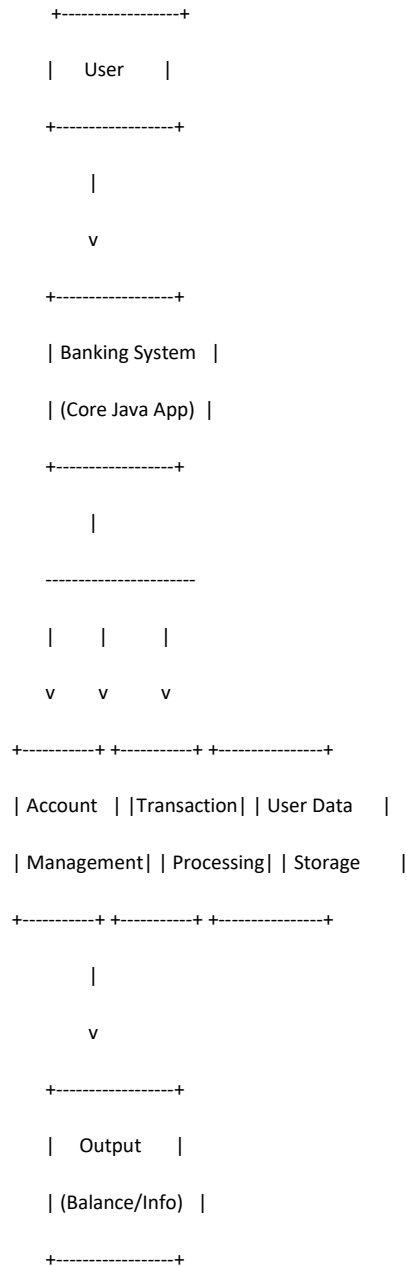
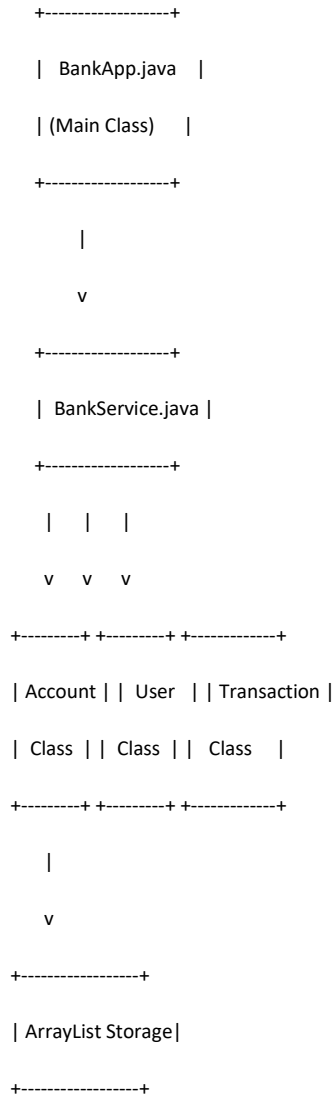


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

```

START
|
v
Display Menu
|
v
User Select Option
|
-----
| | | | |
v v v v v
Register Login Deposit Withdraw Exit
| | | |
v v v v
Process Operation (Validation)
|
v
Show Output
|
v
Repeat / Exit
  
```


6 Performance Test

The performance testing of the Banking Information System was carried out to evaluate how efficiently the system performs under different conditions. This helps in understanding whether the system is suitable for real-world applications.

- ◆ **Identified Constraints:**

The following constraints were considered:

- **Memory Usage:** Efficient storage of user accounts and transactions
- **Execution Speed (MIPS):** Fast processing of operations like deposit, withdrawal
- **Accuracy:** Correct calculation of balance and transactions
- **Scalability:** Ability to handle multiple users (limited in current version)

- ◆ **Handling of Constraints:**

- **Memory:**
Used Java **ArrayList** to store data dynamically, reducing unnecessary memory usage.
- **Speed:**
Optimized logic with simple conditional checks and efficient methods.
- **Accuracy:**
Implemented proper validation (e.g., insufficient balance check before withdrawal).
- **Scalability:**
Modular class design (Account, Transaction, BankService) allows future expansion.

- ◆ **Limitations:**

- Data is stored in memory (not persistent)
- Not tested with very large datasets
- Console-based system limits real-world scalability

- ♦ **Recommendations:**
- Use **MySQL database** for persistent storage
- Implement **multithreading** for handling multiple users
- Add **GUI/Web interface** for better usability
- Optimize algorithms for large-scale systems

6.1 Test Plan/ Test Cases

Test Case ID	Test Scenario	Input	Expected Output	Result
TC01	User Registration	Valid username	Account created successfully	Pass
TC02	Login	Correct credentials	Login successful	Pass
TC03	Deposit Money	Amount = 1000	Balance updated correctly	Pass
TC04	Withdraw Money	Amount = 500	Balance deducted	Pass
TC05	Withdraw (Low Balance)	Amount > Balance	Error message displayed	Pass
TC06	Fund Transfer	Valid accounts	Amount transferred successfully	Pass
TC07	View Statement	Transaction data	Correct transaction history shown	Pass

6.2 Test Procedure

The following procedure was used to test the system:

1. Run the application using Java (VS Code/IntelliJ)
2. Register a new user
3. Login using credentials
4. Perform operations:
 - Deposit
 - Withdraw
 - Fund Transfer
5. Verify outputs after each operation
6. Check for error handling (invalid input, low balance)
7. Repeat tests multiple times to ensure consistency

6.3 Performance Outcome

The system performed efficiently under normal conditions.

7 ✓ Results:

- Fast execution of operations
- Accurate balance calculations
- Stable performance without crashes
- Proper error handling

8 ✓ Overall:

The system meets basic performance requirements for a small-scale banking application. It demonstrates how real-world systems can be designed using Core Java concepts.

9 My learnings

During this six-week internship, I gained valuable technical and practical knowledge that will significantly help in my career growth.

I learned **Core Java programming** in depth, including important concepts such as Object-Oriented Programming (OOP), classes and objects, exception handling, and collections like ArrayList. Through the development of the **Banking Information System**, I understood how real-world applications are designed and implemented.

I also improved my **problem-solving skills** by handling different scenarios such as transaction validation, error handling, and data management. Working on this project helped me understand the importance of writing clean, modular, and maintainable code.

Additionally, I learned how to:

- Structure a complete project
- Debug and test applications
- Use tools like GitHub for version control
- Document a project professionally

This internship enhanced my **logical thinking, coding efficiency, and confidence**, which will be very useful for future software development roles and technical interviews.

10 Future work scope

Although the current system successfully performs basic banking operations, there are several enhancements that can be implemented in the future.

Some possible improvements are:

- **Graphical User Interface (GUI):**
Develop a user-friendly interface using Java Swing or JavaFX instead of a console-based system.
- **Database Integration:**
Use MySQL or other databases to store user and transaction data permanently.
- **Authentication & Security:**
Implement advanced security features like password encryption and OTP-based login.
- **Web-Based Application:**
Convert the system into a web application using modern technologies like Spring Boot and React.
- **Multi-user Support:**
Enable concurrent access using multithreading.
- **Advanced Features:**
Add features like loan management, interest calculation, and account statements in PDF format.

Due to time limitations during the internship, these features could not be implemented, but they can be developed in future versions to make the system more powerful and industry-ready.

